

Is a Hot Dog a Sandwich? A Data Science Approach

Analyzing Culinary Definitions & Public Perception



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Motivation

- There are many conflicting sources about the claim that hot dogs are sandwiches.
 - The USDA, Merriam Webster Dictionary, and even Oscar Mayer themselves claim that Hot Dogs are sandwiches
 - The National Hot Dog & Sausage Council and Joey Chestnut state that they are not sandwiches
- While many differing opinions have been expressed on a national level, we aimed to find a consensus for how the UVA Full Time Undergraduate population feels about hot dog classification



Hypothesis

- Less than half of UVA students believe that a hot dog is a sandwich.
- Null Hypothesis: The true proportion of hot dog-is-a-sandwich believers at uva is 0.5
- Alternate Hypothesis: The true proportion of students at uva who believe a hot dog is a sandwich is less than 0.5



Data

Total students at UVA: ~17,000

Survey Responses: 79

Undergrad Responses: 76

Final Cohort

The survey collected the following data:

- Undergraduate student status (yes/no)
- Year of Study (1st, 2nd, 3rd, or 4th)
- Is a hotdog a sandwich (yes/no)

Please note the following potential biases:

- Response bias - responses were voluntary
- Selection bias - responders were invited by the survey creators

Analysis Plan

Pre-Analysis: Data Cleaning - Filter out non UVA undergraduate students

Step 1: One-tailed T-test our hypothesis against a p value of 0.05

Step 2: View breakdown overall and by year via circle graphs

Step 3: View raw counts by year via bar graph

Step 4: Spearman analysis & chi-squared test to find correlation between student year and hot dog belief

Results

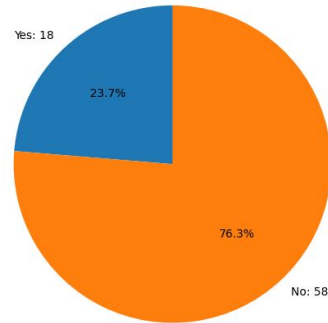
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Number of UVA students (n): 76  
Number saying 'Yes': 18  
Sample proportion  $\hat{p}$ : 0.2368  
z statistic: -5.3962  
One-sided p-value ( $p < 0.5$ ): 3.40371e-08  
Two-sided p-value: 6.80741e-08
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The one-sided p-value is less than 0.05, confirming the alternate hypothesis that the true proportion of students who believe a hot dot is a sandwich is less than 0.5

*note: a two sided t-test was also performed just for comparison

Results (2)

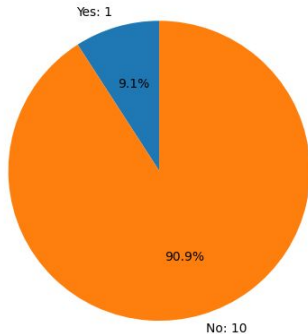
Is a hotdog a sandwich? (Overall, UVA only)



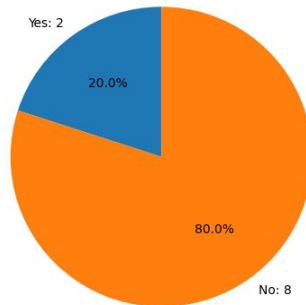
The majority of students in all years believe a hot dog is not a sandwich.

However, the proportion of students saying it is a sandwich increase with each year.

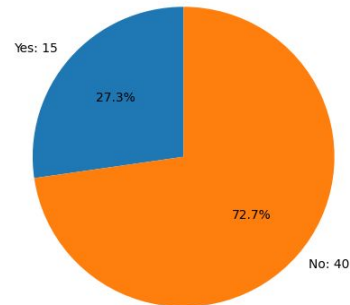
2nd year (UVA only)



3rd year (UVA only)

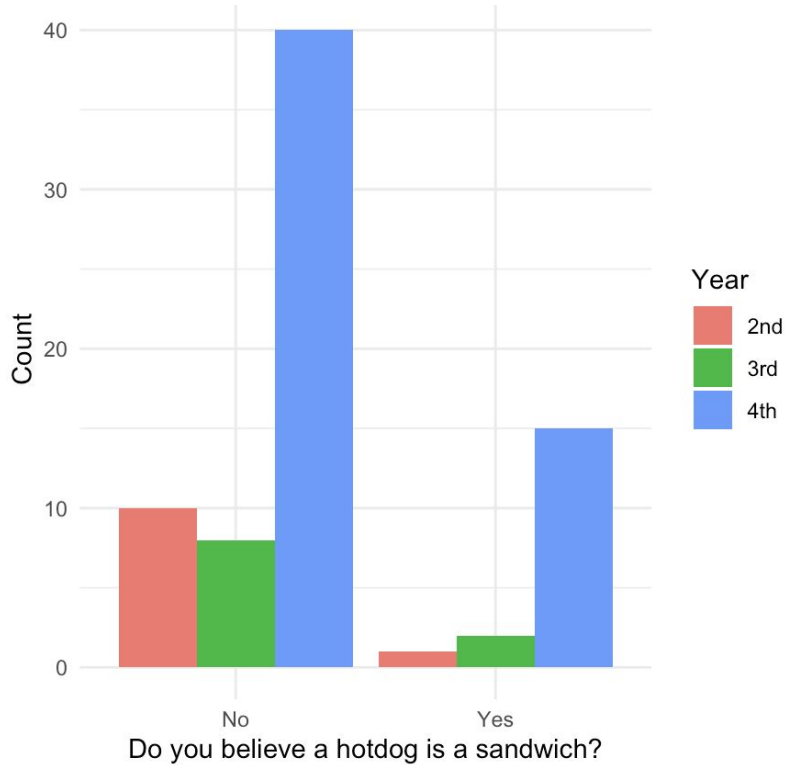


4th year (UVA only)



*Note: there were no 1st year responses to the survey

Results (3)



The majority of our data comes from 4th year students.

Results (4)

Spearman Rank Correlation Results (UVA Only)

Rho: 0.146

P-value: 0.2094

Chi-Square Test Results (UVA Only)

Chi-square statistic: 1.763

P-value: 0.4142

Degrees of freedom: 2

With high p-values, neither the Spearman correlation nor the chi-square test provides evidence of a statistically significant association between year in school and belief that a hot dog is a sandwich.

Next Steps

- Larger and Random sample size to eliminate bias and get a more representative model
- Ask more questions to extend our data analysis to identify predictive factors
- Survey other schools and populations for comparative purposes
- Extend survey classification choices (e.g. Taco)

